2

On Radar Cross Section of Antennas in General

2.1 INTRODUCTION

It is well known that the RCS of any antenna can be significantly reduced by placing a suitably shaped bandpass radome in front of it [35]. When the radome is opaque, the incident signal will primarily be reflected in the specular direction while the backscattered signal will be low as illustrated in Fig. 2.1. However, when the radome is transparent, no significant reduction of the antenna RCS will take place. Thus, the observable RCS will depend primarily on the antenna per se and whatever is behind the radome—for example, the back wall. It therefore becomes important to ask the scientifically very interesting question, "Is it possible to design an antenna that basically is invisible over a broad band in the backward sector without sacrificing its efficiency?"

Most readers will say no. They typically base their answer on well-documented facts about the most commonly used antennas such as a single dipole or monopole, the horn, the flat spiral, the corner reflector, the polyrod, the patch, the log periodic, the helical with a groundplane, and many more. These are all lacking in their ability to produce a low RCS over a broad frequency range when properly matched. Thus, we shall in this chapter instead concentrate on one of the few concepts that can truly produce invisibility in the backward sector, namely the large flat aperture in the form of an array backed by a groundplane and with uniform aperture illumination. The tapered case will also be discussed; and it will be shown that also in that case, invisibility is conceptually compatible with 100% efficiency.

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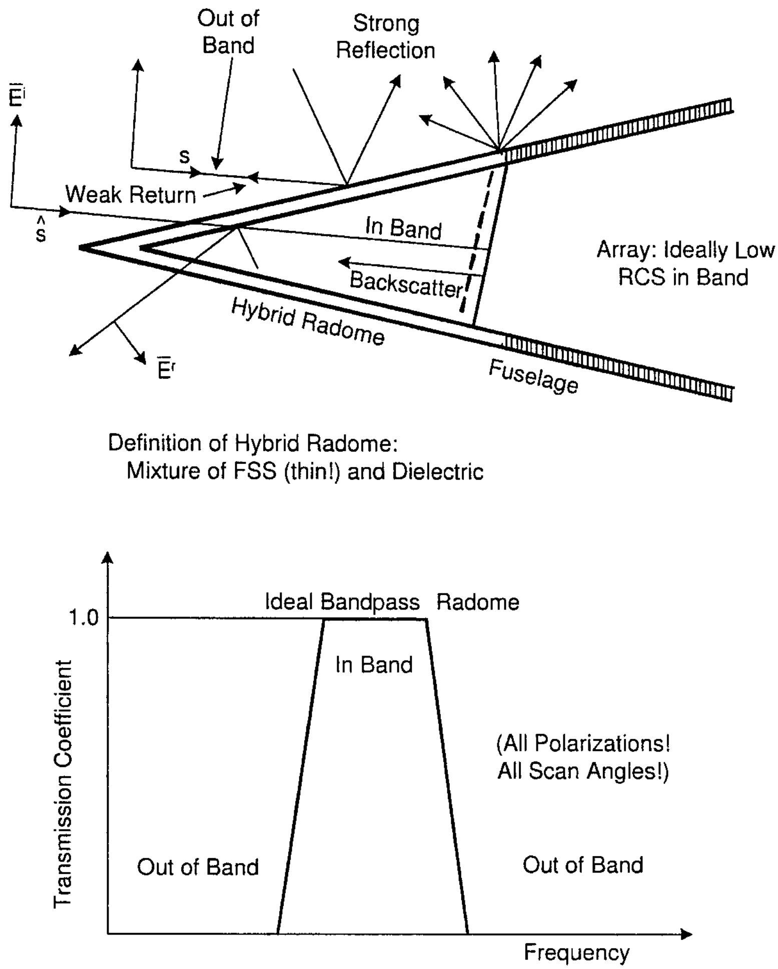


Fig. 2.1 Use of a hybrid radome with bandpass characteristic to reduce the antenna RCS out of band.

A large flat aperture is most often associated with a narrow pencil beam. However, we shall in Chapter 7 consider antennas with omnidirectional pattern and low visibility in the backward direction over a broad band. Further discussed in Chapter 8 is how to design a feed for a parabolic cylinder that will produce an RCS about 6 dB lower than with no feed at all. The design of such a feed is closely related to the omnidirectional design. We emphasize, however, that parabolic systems never can attain the inherently low RCS level encountered for the flat aperture over a broad frequency band.

In the next section we will present the classical fundamentals of antenna RCS valid for all antennas. They are important in order to better understand the intricacies of antenna RCS.

2.2 FUNDAMENTALS OF ANTENNA RCS

In this section we shall show that the field scattered (reradiated) from an antenna is comprised of two components:

1. The **antenna mode** component depending on the gain G, the load impedance Z_L, the polarization, the angle of incidence and the frequency.
2. The **residual mode** component representing whatever must be added to the antenna mode component to obtain the total RCS. It may or may not depend on the gain G, the polarization, the frequency, and the angle of incidence, but never on the load impedance Z_L.

The definition of the residual mode admittedly violates a fundamental scientific principle: Never explain something unknown by something else unknown. Nevertheless, the concept is extremely useful for understanding antenna scattering.

Thus, we shall show several important examples where the residual scattering component is determined and illustrate how it relates to the antenna mode component.

An alternative and somewhat older nomenclature for the residual component is the structural scattering. We do not particularly recommend this nomenclature because it is somewhat misleading. For example, large arrays of dipoles without a groundplane will be shown later (see Section 2.6) to have as much residual scattering as antenna mode scattering. However, when we add a groundplane (yes, add more structure) the antenna mode is increased by a factor of four while the residual scattering simply as shown later is equal to zero. Thus, we avoid the word "structure" because it is burdened by the misconception that more structure means more scattering due to the structure. It may or it may not. See also Sections 2.14.1 and 2.14.2.

2.2.1 The Antenna Mode

More specifically, let us now consider an antenna exposed to an incident plane wave propagating in the direction $\hat{s}_i$ and with power density Φ_i as shown in Fig. 2.2. If the load impedance Z_L is conjugate-matched to the antenna impedance Z_A, the received power is maximum and given by [36]

$$P_{rec}^{i} = \frac{\lambda^2}{4\pi} G_i \Phi_i p_i, \tag{2.1}$$

where G_i is the antenna gain in the direction of incidence $\hat{s}_i$ and p_i is the polarization mismatch factor between the incident E-field and the polarization of the antenna, that is, $0 \leq p_i \leq 1$.

We next seek the reflection coefficient Γ between the antenna impedance Z_A and Z_L. If one of them is real, the usual simple expression for Γ is valid (see later). However, if they are both complex as shown in Fig. 2.3, top, we

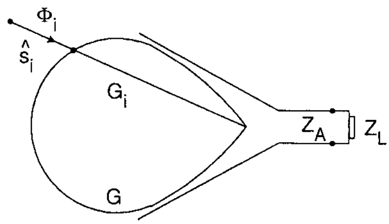


Fig. 2.2 *An antenna with gain G_i being exposed to an incident plane wave with power density Φ_i and direction $\hat{s}_i$ will receive the power $P^i_{rec} = \frac{\lambda^2}{4\pi} G_i \Phi_i p_i$, where p_i represent the polarization mismatch between the antenna and the plane wave.*

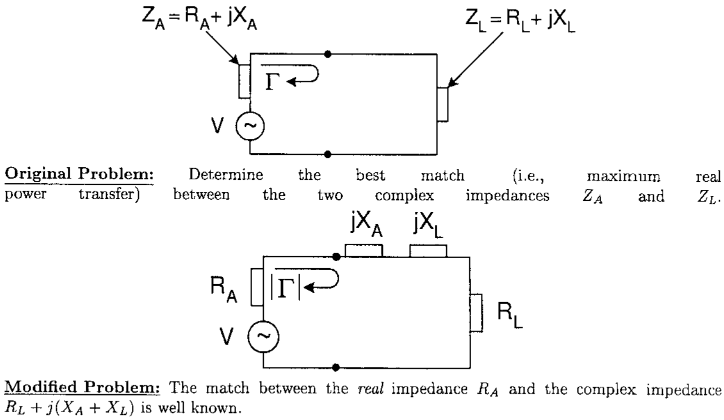


Fig. 2.3 *How the reflection coefficient Γ between two complex impedance $Z_A = R_A + jX_A$ and $Z_L = R_L + jX_L$ is reduced to the simpler problem between the real impedance R_A and the complex load $R_L + j(X_A + X_L)$.*

must modify the original problem to the one shown in Fig. 2.3, bottom. This is based on the simple fact that for a two-port lossless circuit the magnitude (not the complex value) of Γ will remain the same no matter where we place the terminals in the circuit. From the modified circuit the magnitude of the reflection coefficient between the real impedance R_A and the complex impedance $R_L + j(X_L + X_A)$ is readily obtained as

$$|\Gamma| = \left| \frac{R_L + j(X_L + X_A) - R_A}{R_L + j(X_L + X_A) + R_A} \right|. \tag{2.2}$$

The received power P_{rec} given by (2.1) will be partly absorbed by the load impedance Z_L and partly reflected back toward the antenna as reflected power given by

$$P_{refl} = P_{rec} |\Gamma|^2, \tag{2.3}$$

where $|\Gamma|$ is given by (2.2).

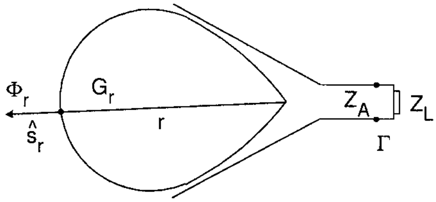


Fig. 2.4 The power P_{rec} received by the antenna will be reradiated as $P_{refl} = P_{rec}|\Gamma|^2$. In the bistatic direction $\hat{s}_r$ the antenna gain is G_r and we obtain a power density $\Phi_r = P_{refl}/4\pi r^2 G_r p_r$, where p_r is the polarization mismatch between the antenna and a receiving antenna.

The reflected power P_{refl} will now be radiated (or rather reradiated) by the antenna like any other signal impressed at its terminals.

If the antenna had been isotropic, the power P_{refl} would at a distance r be uniformly distributed over a sphere with surface area $4\pi r^2$. Thus, the power density at distance r would be $P_{refl}/4\pi r^2$. However, if the antenna has a gain G_r in the direction of radiation $\hat{s}_r$ as shown in Fig. 2.4, the power density in that direction and at distance r will be

$$\Phi_r = \frac{P_{refl}}{4\pi r^2} G_r p_r, \tag{2.4}$$

where p_r is the polarization mismatch factor between the scattering antenna and a receiving antenna located in the far field, that is, $0 \le p_r \le 1$.

Substituting (2.1) and (2.3) in (2.4), we obtain

$$\Phi_r = \frac{\lambda^2}{16\pi^2 r^2} G_i G_r p_i p_r |\Gamma|^2 \Phi_i. \tag{2.5}$$

The definition of the radar cross section (RCS) is illustrated in Fig. 2.5. Here a fictitious flat plate, with area σ_{ant}, intercepts an incident plane wave with power density Φ_i—that is, the intercepted power is $\sigma_{ant}\Phi_i$.

If this power is spread uniformly in space at distance r, the power density Φ_{ant} is

$$\Phi_{ant} = \frac{\sigma_{ant}\Phi_i}{4\pi r^2},$$

or

$$\sigma_{ant} = 4\pi r^2 \frac{\Phi_{ant}}{\Phi_i}. \tag{2.6}$$

The radar cross section of the antenna mode component is now defined as an area σ_{ant} so large that the power density Φ_{ant} associated with the fictitious plate is the same as the power density Φ_r associated with the antenna mode component; that is, we set

$$\Phi_{ant} = \Phi_r. \tag{2.7}$$

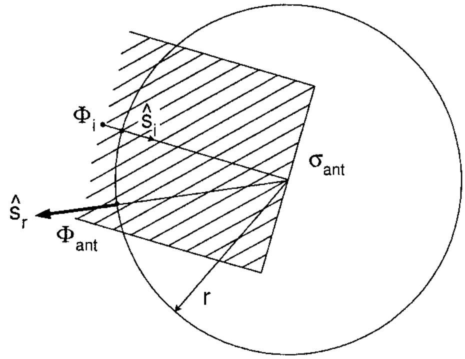


Fig. 2.5 The power intercepted by a fictitious flat plate with area σ_{ant} is $\sigma_{ant}\Phi_i$. When spread uniformly in space, it produces at distance r the power density $\Phi_{ant} = \sigma_{ant}\Phi_i/4\pi r^2$ or $\sigma_{ant} = 4\pi r^2 \Phi_{ant}/\Phi_i$. We define σ_{ant} such that $\Phi_{ant} = \Phi_r$ — that is, equal to the power density produced by the antenna. Thus, from earlier, $\sigma_{ant} = \frac{\lambda^2}{4\pi} G_i G_r p_i p_r |\Gamma|^2$.

Substituting (2.5) and (2.7) into (2.6) yields the radar cross section of the antenna mode:

$$\sigma_{ant} = \frac{\lambda^2}{4\pi} G_i G_r p_i p_r |\Gamma|^2. \tag{2.8}$$

2.2.2 The Residual Mode

Expression (2.8) will in general not constitute the entire RCS of an antenna. In fact, it is only a component of the total radar cross section, which implies that there may be something else. This is usually called the residual (or structural) component σ_{res}. It was defined earlier as "whatever must be added to the field associated with σ_{ant} as given by (2.8) in order to obtain the field associated with the total antenna RCS," that is,

$$\sigma_{tot} \equiv \frac{\lambda^2}{4\pi} G_i G_r p_i p_r |\Gamma + C|^2 \tag{2.9}$$

and the Residual scattering cross section is defined as

$$\sigma_{res} \equiv \frac{\lambda^2}{4\pi} G_i G_r p_i p_r |C|^2. \tag{2.10}$$

Note: $\sigma_{tot} \neq \sigma_{ant} + \sigma_{res}$ in general.

Inspection of (2.10) might seem to imply that σ_{res} is proportional to $G_i G_r p_i p_r$. This is not necessarily the case. In general it is a more intricate function. In fact, the only reason we write it on the form in (2.10) is that it enables us to add

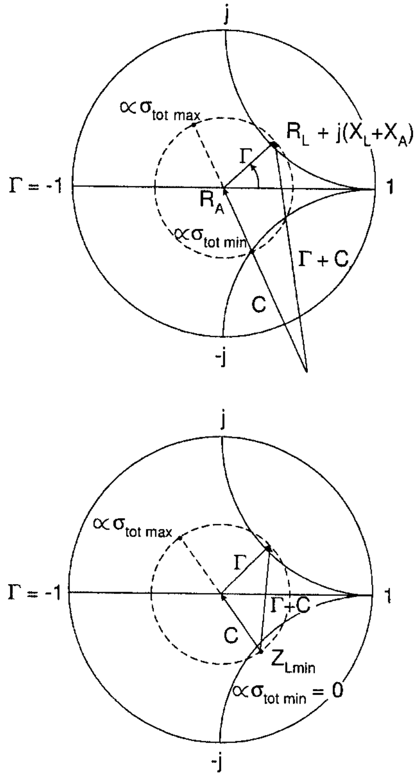


Fig. 2.6 Antenna mode: $\sigma_{ant} = \frac{\lambda^2}{4\pi}G^2|\Gamma|^2$. Residual mode: $\sigma_{res} = \frac{\lambda^2}{4\pi}G^2|C|^2$. Total antenna RCS: $\sigma_{tot} = \frac{\lambda^2}{4\pi}G^2|\Gamma + C|^2$. All antenna RCS can be envisioned as made up of the antenna mode component σ_{ant} and the residual component σ_{res}. The antenna mode component is proportional to $|\Gamma|$ and can therefore be depicted in a Smith chart. The residual component represented by C can be larger than Γ (top) or smaller (bottom).

it to σ_{ant} to obtain σ_{tot} as given by (2.9). However, it does not depend on Z_L. Otherwise it would have been part of σ_{ant}.

More specifically, consider Fig. 2.6, top. We show here a Smith chart normalized to R_A. Recall further that one of the unique features of the Smith chart is that the reflection coefficient Γ as given by (2.2) is a phasor measured from the center R_A of the Smith chart to the modified load impedance $R_L + j(X_L + X_A)$ as shown in the Smith chart. The magnitude of the field associated with the antenna mode component is proportional to $|\Gamma|$ [see (2.8)], while the magnitude of the field associated with the residual mode component is proportional to $|C|$ [see (2.10)]. Thus, according to (2.9) the field associated with σ_{tot} is proportional to the magnitude of the phasor sum $(\Gamma + C)$ as also shown in the same Smith chart.

The beauty of this representation is that it clearly shows how σ_{tot} varies with the load impedance Z_L (as mentioned above, C does not depend on Z_L). For example, we may let Z_L vary in such a way that the VSWR is constant. In that case the tip of Γ will move along a circle with radius $|\Gamma| \leq 1$ as shown in Fig. 2.6, top. We can in that way obtain a maximum $\sigma_{tot\ max}$ as well as a minimum $\sigma_{tot\ min}$ as shown. For conjugate match $Z_L = Z_A^*$ and $\Gamma = 0$; that is, σ_{tot} is proportional to $|C|^2$. Thus, for matching the load to maximum power transfer, σ_{tot} can be substantial if $|C|$ is large; and what is worse, if $|C| > |\Gamma|$ as is the case in Fig. 2.6, top, we can never obtain $\sigma_{tot} = 0$ for any load.

2.3 HOW TO OBTAIN A LOW σ_{tot} BY CANCELLATION (NOT RECOMMENDED)

We considered above the case where the residual component C was bigger than Γ. We found that σ_{tot} could never become zero no matter how we chose our load impedance Z_L. However, if we instead considered an antenna where $|C| < 1$ as shown in Fig. 2.6, bottom (there are no particular limitations on C), we readily observe that we can choose the load impedance Z_L such that $C + \Gamma = 0$; that is, $\sigma_{tot\ min} = 0$. This technique is referred to as RCS control by cancellation.

There are two strikes against this approach. First of all the load impedance Z_L is not necessarily adjusted for conjugate match; that is, the power transfer is not perfect. But the biggest flaw is that the cancellation is in general very frequency-sensitive—that is, narrowbanded (Z_L and Z_A will in general change significantly and differently with frequency). And as if that is not enough, the cancellation condition changes in general with angle of incidence and polarization.

Therefore this approach should in general be discarded as being primarily of academic interest. See also Sections 2.14.1, 2.14.4, and 2.14.7, as well as Section 2.9 and Problem 2.2.

2.4 HOW DO WE OBTAIN LOW σ_{tot} OVER A BROAD BAND?

The answer to that question should by now be fairly obvious: Choose an antenna with a residual scattering close to zero (i.e., $C \sim 0$) and keep Γ as low as possible over as a broad a band as possible as illustrated in Fig. 2.7. That will ensure maximum power transfer (or almost) and low σ_{tot} at the same time.

The reaction to this suggestion is typically something like: Well, we have measured dipoles, horns, parabolic dishes, flat spirals as well as helical antennas with groundplane, and what not, and we have never come across an antenna with no residual scattering. In fact we are not even sure whether an antenna without residual scattering violates certain fundamental rules!

Actually, they are almost correct, but it is bad science to generalize based on a limited number of cases. The fact is that antenna configurations without residual scattering do not violate any fundamental law and that they do indeed exist. These will be discussed in Sections 2.6 and 2.7. However, let us first pay homage to some of the key people who pioneered the theory about antenna scattering.

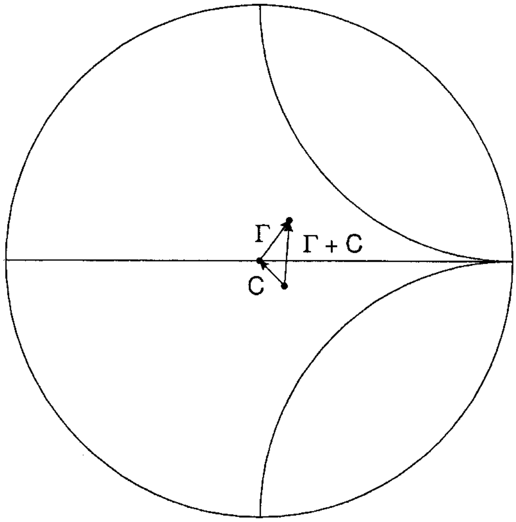


Fig. 2.7 To obtain a low RCS over a broad band an antenna ideally should have (a) as low a residual scattering as possible (i.e., C $\sim$ 0) and (b) a broadband match yielding $\Gamma \sim 0$.

2.5 A LITTLE HISTORY

When the author joined the group of Professor Leon Peters, Jr., at the Ohio State University Antenna Laboratory (currently ElectroScience Laboratory) in the mid-1960s as a graduate student, he was considered a competent antenna engineer with a good industrial background. He had a good knowledge of antenna impedance and considerable experience in matching and he knew how to obtain a radiation pattern from a complex antenna system. However, he had not even heard about antenna radar cross section. And that was typical for most antenna engineers at that time.

He was first introduced to the mysteries of antenna scattering during a short course given at Ohio State University in 1966. One of the sessions was devoted to the RCS of antennas [37] and conducted by Professor Robert Garbacz, at that time a graduate student of Professor Edward Kennaugh. Furthermore, some time before in 1963 another graduate student and a close friend of Garbacz, namely Robert B. Green, wrote his dissertation that dealt exclusively with RCS of antennas [38]. It was that work that became the foundation for Section 2.2 in this chapter.

However, it appears that the concept describing how the scattering from an antenna is made up of two components was presented for the first time in an OSU report by McEntee [39]. He clearly recognized that the scattering (reradiation) is coming from the mismatch between antenna and load impedance(s) (the antenna mode), and, in addition, a scattering component comes from somewhere else associated with the antenna (the residual or structural component). He did not actually call these components by these names. Apparently that came later. All of this activity did of course take place under the watchful eyes of Professors Peters and Kennaugh, who provided many of the basic ideas and concepts.

However, there were other important contributions. Hansen discussed the relationship between antennas as scatterers and as radiators [40]. Garbacz discussed measurement techniques [41], and so did Appel-Hansen [42], Wang et al. [43] and King [44]. Several oral papers about measurements were presented by Heidrich and Wiesbeck [45–48], culminating with the dissertation by Heidrich [49].

2.6 ON THE RCS OF ARRAYS

In the world of antennas, arrays of wires or dipoles have a unique position. They can be designed to have very large bandwidth (>7:1 with VSWR <2); see Chapter 6. They can be designed to have very low radar return, possibly the only concept available at this point in time. Finally they constitute a periodic structure and are therefore properly discussed in this book.

In many respects it makes a significant difference whether the array has a groundplane or not. Thus we shall look in detail at each of these cases separately.

Note that the discussion to follow serves merely the purpose of explaining the physics of antenna scattering. Actual rigorous calculations are obtained from either the PMM or the SPLAT program. Numerous rigorous calculated examples are given in Chapter 5. It has been the author's experience that while most scientists can use a computer program, only a minority understand "what goes on" and are lacking in design ability. Thus, although our presentation is based on approximate ideas, they are very important to the serious researcher.

2.6.1 Arrays of Dipoles without a Groundplane

Basically we will consider arrays of infinite extent in both the X and Z directions. However, the fundamental concept discussed here will also apply to very large arrays (in terms of wavelength) as long as we disregard edge effects. These will be discussed later in Chapter 5 and focus on the variation of the element impedance.

In Fig. 2.8 we show a dipole array seen edge-on under three different load conditions. The element lengths are $\sim\lambda/2$. In the case to the left the terminals are open-circuited. Thus, looking into the antenna terminals we observe a reflection coefficient $\Gamma = +1$ as also shown in the Smith chart underneath the array.

However, we can also consider such a structure as a frequency selective surface. When exposed to an incident plane wave the reflection coefficient $\Gamma_{FSS} \sim 0$. (From this viewpoint we have assumed here that the total scattering from $\lambda/4$ segments is negligible compared to that from $\lambda/2$ dipoles. While this is an approximation depending on wire radius, it does not alter the fundamental approach.)

$\Gamma_{FSS} \sim 0$ simply implies that $\sigma_{tot} \sim 0$, and from (2.9) we then obtain

$$C \sim -\Gamma = -1 \tag{2.11}$$

as also indicated in the Smith chart below the array.

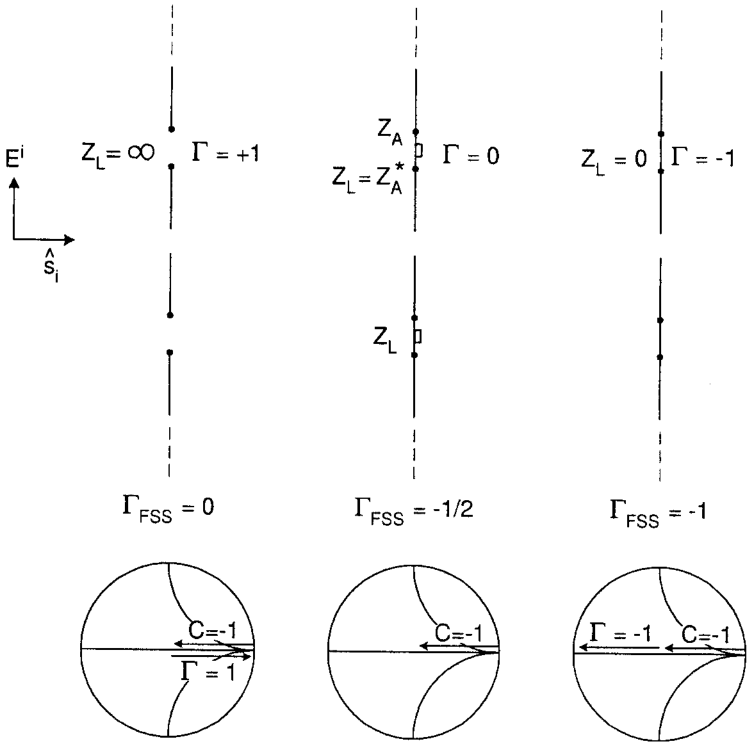


Fig. 2.8 A large array without a groundplane seen edge-on for three load conditions. Left: The terminals are open-circuited, that is, $\Gamma = +1$. Middle: The terminals are loaded with a conjugate match, that is, $\Gamma = 0$. Right: the terminals are short-circuited, that is, $\Gamma = -1$.

Obviously, (2.11) supports the statement made several times earlier (see Section 2.2), namely that an array without a groundplane has about as much residual as antenna mode scattering.

Next we consider the case where Z_L is conjugate-matched to the antenna impedance Z_A. Thus, the reflection coefficient at the array terminals is $\Gamma = 0$, and the backscatter from the array is simply proportional to $C = -1$.

Finally we consider the case to the right in Fig. 2.8. We have here loaded the array at its terminals with short circuits; thus $\Gamma = -1$ as indicated in the Smith chart underneath the array. The total backscattered field is proportional to $\Gamma + C \sim -1 - 1 = -2$ [see (2.9)]. However, when viewed as an FSS of elements with length $2l \sim \lambda/2$, we also know that such a surface reflects as a groundplane. Thus the reflection coefficient for an incident wave is $\Gamma_{FSS} = -1$. In other words, $\Gamma + C = -2$ produces a reflection coefficient equal to $\Gamma_{FSS} = -1$, and consequently the matched case in the middle with $C = -1$ will produce a reflection coefficient equal to $\Gamma_{FSS} = -1/2$.

The observation that loading an array with load impedances $Z_L = Z_A^*$ produces a backscattered radar cross section 6 dB below the backscatter obtained for the short-circuited case is often referred to as the "6-dB Rule." It is characteristic for a class of antennas usually referred to as minimum scattering antennas (MSA) (according to the classical definition as will be discussed in Section 2.10). It is often falsely assumed that this rule pertains to all antennas. It does not. In fact, we shall in the next section consider arrays with a groundplane and observe that the 6-dB Rule fails completely as it does for many of the most important antenna configurations. And thank heaven for that! (See also Problem 2.3.)

2.6.2 Arrays of Dipoles Backed by a Groundplane

We now add a groundplane to our array of dipoles as shown in Fig. 2.9. This will distinctly change the antenna impedance Z_A, but that is of no particular concern for our present purpose.

An incident plane wave with direction of propagation $\hat{s}_i$ will induce currents on the elements as discussed in reference 50. They in turn will radiate a spectrum of plane inhomogeneous waves. The zero-order mode will always propagate in the specular direction $\hat{s}_r$ and is denoted by "1" in Fig. 2.9. Similarly, another plane wave will be radiated symmetrically to "1" and then be reflected in the

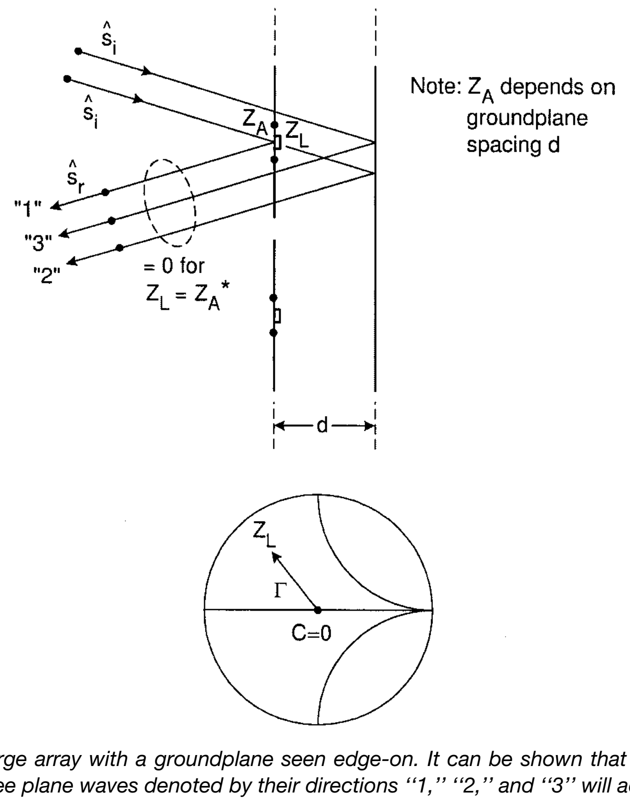


Fig. 2.9 A large array with a groundplane seen edge-on. It can be shown that for conjugate match the three plane waves denoted by their directions "1," "2," and "3" will add up to zero; that is, everything is absorbed.

groundplane such that it also propagates in the specular direction. It is denoted "2." Finally, the incident field will go straight through the elements and be reflected as "3" (remember in our model we substitute the conducting elements for just electric currents. These are always "transparent).”

We now observe the scattered field at a distance so large that all the evanescent waves have practically died out (typically that would be any distance from the dipoles exceeding about $\lambda/4$ unless the interelement spacings D_x and D_z are large). We also assume D_x and D_z to be so small that no grating lobes can exist. It is now left as an exercise for the students to show (see Problem 2.4) that if the load impedance Z_L is conjugate-matched to Z_A the sum of the three plane waves "1," "2," and "3" will add up to zero (we assume that our point of observation is beyond the extent of the evanescent waves). Thus, the incident field must be totally absorbed since nothing is reflected for conjugate match; that is, the residual scattering is zero, or $C = 0$. Obviously this constitutes the ideal situation as illustrated in Fig. 2.7: If the antenna is conjugate-matched, $\Gamma = 0$ and the antenna is invisible in the backward sector.

This result is surprising to many readers. However, it has actually been known by implication for at least half a century. As any classical textbook about antennas will tell you [51], the receiving area of a large array with uniform aperture distribution and no groundplane is half its physical size. Thus, when conjugate-matched, such an array will receive half the power incident upon it. The other half (i.e., down 3 dB) will be radiated equally in the forward and backward directions—that is, down another 3 dB. Thus we have simply verified the "6-dB Rule" introduced earlier.

Furthermore, it is also well known that if a large array is provided with a groundplane, the receiving area is simply equal to the physical area [52]. In other words, an array with a groundplane and conjugate match will receive *all* the energy incident upon it and will consequently not scatter any energy in the backward direction (*Remember*: Only for uniform aperture distribution. The tapered aperture distribution is discussed in Section 2.11.2).

As an antenna engineer these facts were well known to the author early on in his work with the RCS of antennas. In fact, they quickly became the guiding light in the pursuit of the invisible antenna concept. In all research it is a tremendous help if you know the final result ahead of time!

2.7 AN ALTERNATIVE APPROACH: THE EQUIVALENT CIRCUIT

The approach taken above is a good illustration of the scattering theory valid for all antennas. However, when we work with large flat arrays as above, it is actually considerably easier to simply consider an equivalent circuit. Note, however, that the primary purpose of such a circuit is to gain understanding and guidance in the design of arrays. It is never used to obtain actual numerical results. These are obtained from the PMM or similar programs.

The equivalent circuit was developed earlier [53] and is actually used extensively in Chapter 6 of this book for understanding large arrays designed for a

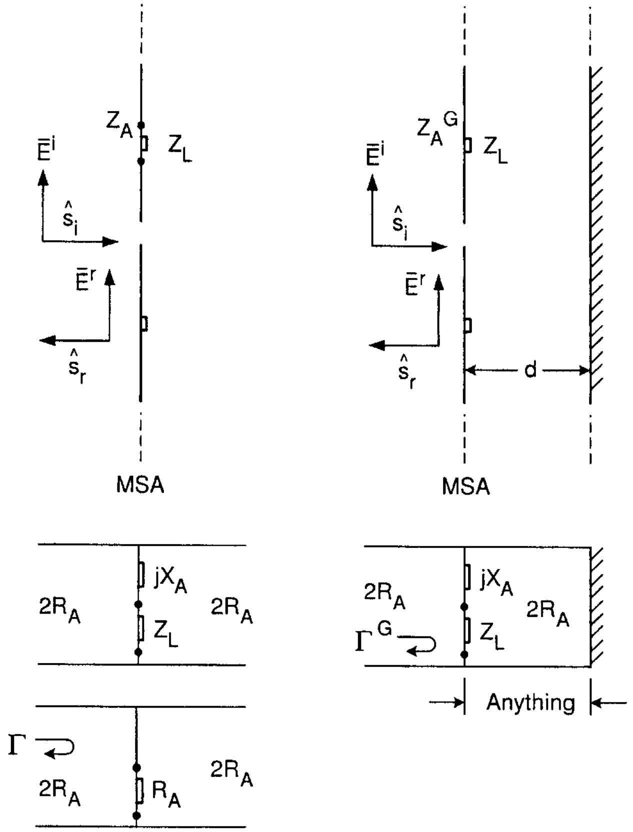


Fig. 2.10 Left: A plane wave is incident upon a large array without a groundplane. As seen from the equivalent circuit below, it will have no backscatter only for $Z_L = \infty$; that is, we will receive no power. For Z_L conjugate matched, $\Gamma = -1/2$ [see (2.12)]. Right: Same situation as before but the array has a groundplane. When Z_L is conjugate-matched, the incident wave is completely absorbed; that is, the RCS equals zero. Note: These equivalent circuits are valid only in the principal planes.

broad band. Thus, it suffices here simply to show the equivalent circuit for an array without a groundplane (see Fig. 2.10, left) and similarly an array with groundplane (see Fig. 2.10, right). In the first case we observe that it consists of an infinite transmission line with characteristic impedance $2R_A$ shunted with the antenna reactance jX_A in series with the load impedance Z_L. For conjugate match we obtain the equivalent circuit shown at the bottom. The reflection coefficient for a signal incident from left or right is then

$$\Gamma = \frac{R_A \| 2R_A - 2R_A}{R_A \| 2R_A + 2R_A} = -\frac{1}{2}, \tag{2.12}$$

that is, we observe the "6-dB Rule" as discussed above. See also Problem 2.5.

We now add a groundplane as shown in Fig. 2.10, right. This will change the original antenna impedance to Z_A^G, and the effect of the groundplane is taken into account by a transmission line in parallel with $jX_A^G + Z_L$ and terminated in a short (the groundplane). If the load impedance Z_L is conjugate-matched to whatever impedance is observed at the terminals, all available power will be absorbed. Since the groundplane case to the right is a lossless two-port device connected to Z_L (see Fig. 2.3), we may conclude that a signal incident along the transmission line from the left will also be completely absorbed; that is, $\Gamma^G = 0$ or the residual scattering is zero. In other words, the signal reflected from an array with a groundplane is ∞ dB below the incident signal if conjugate matched. This is sometimes referred to as the "∞-dB Rule." Obviously, the 6-dB Rule does not hold at all in the groundplane case. One substantial difference between the two cases is that the former is a three-port device, whereas the latter is a lossless two-port. Only for $Z_L = \infty$ could the array without groundplane be "invisible;" and since the power received in that case would be zero, it is primarily of academic interest only. See also Section 5.3.

Note that the spacing d between array and groundplane can be anything but $n\lambda/2$. See also discussion about this subject in Section 6.12.2.

We finally note that an array with a groundplane and resistive loads actually belongs to a simple form of circuit analog absorbers. This relationship was pointed out in reference 54 and is also discussed in a different context in Section 6.12.1.4.

2.8 ON THE RADIATION FROM INFINITE VERSUS FINITE ARRAYS

2.8.1 Infinite Arrays

It is well known that the field radiated from an infinite array can be written as a sum of a finite number of propagating waves (the principal term corresponding to $k, n = 0, 0$ plus grating waves if any) and an infinite number of evanescent waves [50]. An example is shown in Fig. 2.11, top, where the direction of the principal propagating wave is denoted $\hat{r}(0, 0)$ and where no grating waves are encountered. Also indicated are the evanescent waves. They will typically die out as the point of observation moves beyond a distance $\sim\lambda/4$ from the array at which point only the propagating waves will exist. Thus, when dealing with infinite arrays we will always be in the "near field" as far as the radiation pattern is concerned; the concept "far field" simply does not exist for an infinite array.

2.8.2 Finite Array

The field radiated from a finite array is shown quantitatively in Fig. 2.11, bottom. Close to the array the field looks approximately as for the infinite array except for some ripples. These are merely caused by the same type of surface waves encountered in Chapter 1 and discussed in more detail in Chapters 4 and 5. They typically occur every time a general aperture is finite rather than infinite.

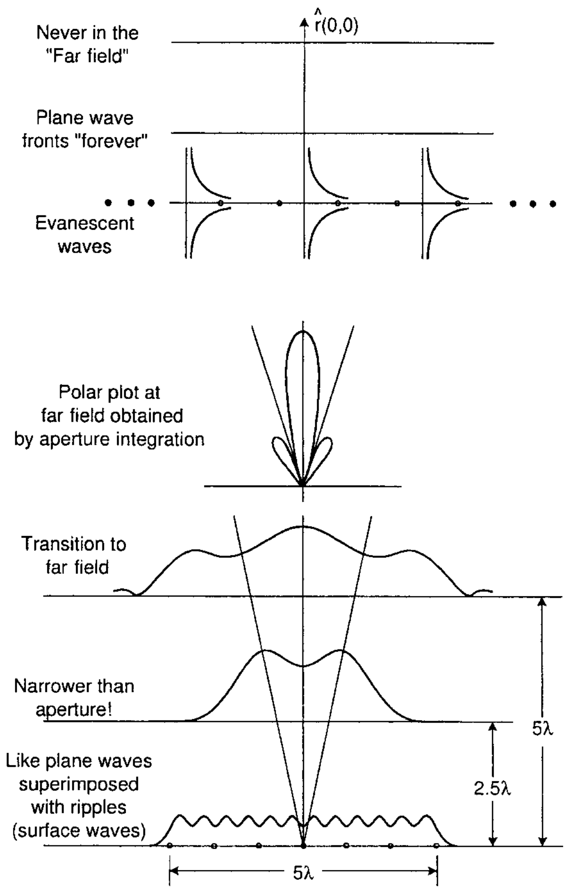


Fig. 2.11 Top: The field from an infinite array consists of a finite number of propagating plane waves and an infinite number of evanescent waves. The sum of the evanescent waves essentially constitutes the near field of the array. Bottom: The field from a finite array at various distances as shown.

As we move away from the finite array to a distance about half the size of the aperture we observe that the extent of the field becomes narrower than the aperture. And at a distance about the size of the aperture it has an extent about the size of the aperture. These observations are typical.

Finally, we show a polar plot of the far field obtained by simple aperture integration. Comparing the infinite and finite cases we observe that the former has an "infinite narrow" beam (like a Dirac-Delta function) in the direction $\hat{r}(0, 0)$ while the latter exhibits the well known far field pattern. At the array elements they both look similar except that the finite case shows ripples along the aperture as caused by the presence of surface waves as discussed in Chapters 1, 4 and 5.

We emphasize again: Fig. 2.11 is merely meant to be qualitative, not quantitative.

2.9 ON TRANSMITTING, RECEIVING, AND SCATTERING RADIATION PATTERN OF FINITE ARRAYS

In the last section we introduced the radiation pattern concept for finite arrays. We shall in this section consider in more detail what happens under transmitting versus receiving or scattering conditions.

Consider a finite array being fed from a single pair of terminals via some kind of harness. If we transmit from those terminals it is a conceptually simple process to obtain a far field transmit pattern Pat^{tr} based on the element currents under transmitting condition.

If we instead receive an incident signal at the same pair of terminals as above and rotate the antenna around its center we will record the receiving pattern Pat^{rec}. From the law of reciprocity we know that the transmit and receiving pattern will always be identical, i.e.

$$Pat^{tr} \equiv Pat^{rec}. \tag{2.13}$$

However, as also stated in the law of reciprocity, this fact does by no means imply that the currents under transmit and receiving conditions are identical. In fact, while they in some cases may be similar, they will in general be different.

Furthermore, if we evaluate the radiation pattern based on the receiving (i.e., scattering) currents, we will obtain the so-called scattering pattern Pat^{scat}. Obviously, based on our observation above concerning currents under receiving conditions, we can state that Pat^{scat} in some cases may be similar to $Pat^{tr} \equiv Pat^{rec}$, but it will in general be different, that is,

$$Pat^{scat} \neq Pat^{tr} \equiv Pat^{rec}. \tag{2.14}$$

We will now illustrate these statements by various examples.

2.9.1 Example I: Large Dipole Array without Groundplane

A large array of dipoles without a groundplane is shown in Fig. 2.12 top. Although not shown in the figure it is assumed that each dipole is fed via a harness containing hybrids as discussed later in Section 2.12. It will then be shown that as far as scattering is concerned, it is equivalent to loading each element with identical load impedances Z_L. And since we at present are most interested in the scattering aspect of such an array, this is quite sufficient for now.

If we impress a voltage at each terminal, we will obtain an element current under transmitting condition. For a total element length equal to $\sim\lambda/2$ or shorter, it will be approximately sinusoidal. However, if we instead expose the array to an incident plane wave we will obtain the element currents under receiving or scattering condition. They are not identical; in fact they are in general different, but if the total element length is $\sim\lambda/2$ or shorter, they will be similar (see also Section 2.11.1 dealing with full-wave dipoles). Now, the transmitting pattern Pat^{tr} is obtained based on the transmitting current. Similarly the scattering pattern

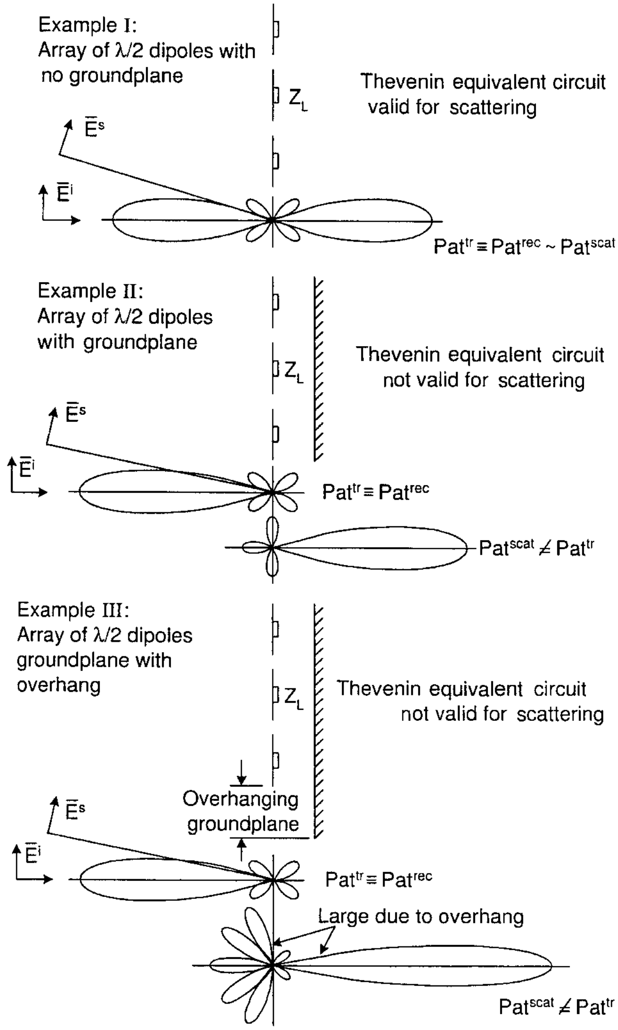


Fig. 2.12 The transmitting, receiving, and scattering pattern. Top: An array of dipoles without a groundplane. Middle: An array of dipoles with a groundplane. Bottom: An array of dipoles with an oversized groundplane.

Pat^{scat} is obtained from the receiving or scattering currents. And since the two current distributions are similar in this case (for element length $\sim\lambda/2$ or shorter), we may conclude that

$$Pat^{tr} \sim Pat^{scat}. \tag{2.15}$$

Note: For the receiving pattern Pat^{rec} we always have from the law of reciprocity $Pat^{rec} \equiv Pat^{tr}$ in spite of the fact that the current distributions in the two cases in general are different.

Based on this observation many assume that the scattering pattern is in general almost the same as the transmitting (or receiving) pattern. However, the next example will illustrate that this is a completely erroneous assumption.

2.9.2 Example II: Large Dipole Array with Groundplane

In Fig. 2.12, middle, we show a large finite array of dipoles backed by a finite groundplane as large as the array. When fed from constant current generators, we obtain the currents under transmitting condition, and the transmit pattern Pat^{tr} is obtained in the traditional way based on these currents and their approximate images in the finite groundplane.

Similarly to Example I above, we shall feed these elements via a harness containing hybrids in order to decouple the elements internally from each other (see Section 2.12). Thus, our scattering model will merely consist of elements loaded with identical load impedances Z_L as shown. We seek the bistatic scattering pattern Pat^{scat} based on the element currents when exposed to an incident plane wave. At this point we shall do so by estimating the scattering pattern in the backward and forward sectors separately (an example of exact calculation is given in Section 5.3).

Let us first examine the backward sector. If we assume that the load impedance Z_L is conjugate-matched to the antenna impedance Z_A, all the energy incident upon this array with a groundplane will (as shown in Section 2.6.2) basically be absorbed. Thus, for this load condition the scattering pattern will simply be given by a number of low-level sidelobes due to the finiteness of the array. For an actual calculated example of what happened when $Z_L \neq Z_A^*$, see Section 5.3.

And now for the forward sector. Right behind the groundplane (i.e., in the forward sector) we realize that the total field to a good approximation is equal to zero over the whole groundplane. We further recall that the total field is given as the sum of the incident field $\overline{E}^i$ and the scattered field $\overline{E}^s$. Thus we readily conclude that $\overline{E}^s \sim -\overline{E}^i$ right behind the groundplane. Furthermore, since the scattering pattern Pat^{scat} is obtained by aperture integration of $\overline{E}^s$ over the back of the groundplane, we may conclude that the scattering pattern in the forward direction is obtained simply as the negative of the incident field $\overline{E}^i$ integrated over the entire groundplane. Thus, the scattering pattern Pat^{scat} has in the forward direction a large mainbeam and in the backward sector some small sidelobes as indicated in Fig. 2.12, middle. In fact, one might be tempted to assume that Pat^{scat} is merely equal to the transmit pattern Pat^{tr} reversed. This is not the case. First of all, recall that the scattering pattern Pat^{scat} in the backward sector is highly dependent on the load impedance Z_L. In fact for Z_L imaginary we will obtain another main beam in the back. This is in sharp contrast to the transmit and receiving pattern that is completely independent of any generator or receiver impedance. Furthermore, while it is true that the main beams for Pat^{tr} and Pat^{scat} for this case are almost identical except for their opposite directions, this will in general not necessarily be the case. The next example will illustrate this point.

2.9.3 Example III: Large Dipole Array with Oversized Groundplane

Let us finally consider an array with the same number of elements as in Examples I and II but where the finite groundplane is somewhat larger than the dipole area as shown in Fig. 2.12, bottom.

The transmit pattern Pat^{tr} will essentially be as for Example II above except for the sidelobe region. Thus, we can immediately conclude that the power received for conjugate match will essentially be the same as for Example II. The energy scattered in the backward sector will, however, be quite different. When conjugate-matched we may think of the scattering area as an absorber (the dipole area) surrounded by a perfectly conducting "ring area," namely the overhanging groundplane. Thus, the scattering in the backward sector is essentially given by the scattering from the ring-shaped area and will consequently consist of many large "sidelobes" with a level depending on the size of the ring-shaped area.

Furthermore, the forward scattering is, as explained earlier, given by integration of the incident field over the entire groundplane—that is, including the overhang. Thus, the forward scattering will essentially consist of a mainbeam that is narrower and larger than the one observed in Example II above.

2.9.4 Final Remarks Concerning Transmitting, Receiving, and Scattering Radiation Pattern of Finite Arrays

Example I illustrated that the scattering pattern Pat^{scat} could essentially have the same shape as the transmit pattern Pat^{tr}, but the amplitude of the former depended on Z_L while the latter was independent of Z_L.

Example II showed that the transmit and scattering pattern in general are quite different and complicated by the fact that the backscatter section is highly dependent on Z_L while the forward sector is basically independent of Z_L.

Examples II and III showed that two antennas can basically have the same transmit pattern but significantly different scattering patterns.

We should finally note that the antenna in Example III could in some cases produce zero backscatter in a few directions by adjusting Z_L. As noted in Section 2.3, this destructive interference method is not recommended.

To many people it is counterintuitive that the transmit and the scattering patterns point in opposite directions. In Problem 2.6 you are asked to find these patterns by direct calculation for a single dipole (or column of dipoles) backed by a single parasitic dipole reflector (or column of parasitic dipoles).

Nothing brings home a point better than simply working it out.

2.10 MINIMUM VERSUS NONMINIMUM SCATTERING ANTENNAS

The *classical definition* of a minimum scattering antenna (MSA) is one that:

1. Scatters as much total power as it absorbs for conjugate match.
2. Has identical transmitting and scattering patterns.

3. Radiates the same fields in the forward and backward directions.
4. Is “invisible” when its terminals are “open-circuited.”

Consequence: Only 6-dB reduction in RCS for conjugate match, see Example I above.

This definition has been known at least since World War II [55] and has been widely quoted eversince [37, 38, 56–58]. What is more, it has often been implied that this class of antenna represents the ultimate as far as low radar cross section is concerned and, consequently, should be treated with great respect!

However, this may not be quite the case (another case of bad notation). In fact, the author suggests a relaxed condition for an MSA, namely one that scatters (in total) only as much as it absorbs—that is, one that is constrained only by condition 1 above (most antennas scatter more in total than they absorb, never less). As we shall see, this relaxed condition leads to a much broader class of antennas, some of which are truly noteworthy. The three examples presented above are excellent illustrations of these concepts. However, let us briefly remind the reader about the Thevenin equivalent circuit as it applies to receiving antennas.

2.10.1 The Thevenin Equivalent Circuit

It is well known that any receiving antenna can be described by a Thevenin equivalent circuit, shown in Fig. 2.13. Here the power delivered to the load impedance Z_L will always correctly represent the power being absorbed by the antenna, while the power being lost in Z_A represents power being reradiated somewhere in space by the antenna. However, the antenna may actually scatter considerably more power than is being lost in Z_A.

2.10.2 Discussion

Let us consider Example I above first. We already established earlier that for conjugate match, half the incident power will be absorbed in Z_L, while the other half will be scattered with one quarter going in the forward direction and another

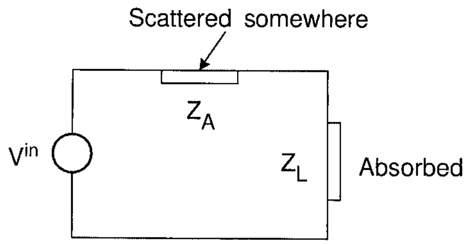


Fig. 2.13 The Thevenin equivalent circuit for a receiving antenna always yields the correct power delivered to the load impedance Z_L. The power lost in the antenna impedance Z_A represents power scattered somewhere, not necessarily in the back direction. Additional scattering not accounted for by Z_A will in general take place.

quarter in the backward sector (i.e., we observe the 6-dB Rule). Furthermore, the transmitting and scattering patterns are basically identical and we obtain as strong a signal in the forward as in the backward direction. Finally, the antenna is basically "invisible" only when $Z_L = \infty$.

Thus, this antenna is truly an MSA in the rigorous classical sense. In addition, the Thevenin equivalent circuit will correctly describe all of these features as far as the array in Example I is concerned.

Similarly, the antenna in Example II will for conjugate match absorb all the energy incident upon it while next to nothing is being scattered in the backward sector. However, as we saw earlier, we will in the forward sector observe a strong scattering pattern obtained by integration of the incident field across the groundplane, which is basically identical to how we obtain the transmitting pattern in the backward direction. Thus, this antenna will scatter as much energy in the forward sector as it absorbs for conjugate match.

However, the transmitting (or receiving) pattern is radically different from the scattering pattern, and it radiates more in the backward than in the forward direction. And when open-circuited an incident field will essentially see just the groundplane and consequently exhibit a very large RCS. Thus, for conjugate match the Thevenin circuit will correctly account for the absorbed and scattered power but fail completely for $Z_L = \infty$. See also Figs. 5.2 and 5.3 for exact calculations.

Thus, this antenna will satisfy only condition 1 above and none of the other three and is therefore not an MSA in the classical sense.

Nevertheless, as we saw earlier, the antenna in Example II can for conjugate match produce a backscatter that is basically ∞ dB below the power scattered when $Z_L = 0$ while the MSA antenna in Example I has a backscatter only 6 dB below the case $Z_L = 0$.

To put it bluntly: The antenna in Example II is the only type that has a "shot" at invisibility when conjugate-matched in spite of the fact that it is not "pedigreed" as is the antenna in Example I.

Finally, the antenna in Example III will, for conjugate match, always scatter more total energy than it absorbs and in addition it will, like the antenna in Example II, not satisfy requirements 2, 3, or 4; that is, it is not an MSA even in the relaxed sense.

2.11 OTHER NONMINIMUM SCATTERING ANTENNAS

We demonstrated above without actual proof that to obtain zero backscatter for an antenna, it should scatter no more total energy than it absorbs when conjugate matched. Very few antennas belong to this elite class in fact most antennas do not. Let us illustrate these statements with some examples.

2.11.1 Large Array of Full-Wave Dipoles

A large array of full-wave dipoles without a groundplane will scatter approximately as much power as it receives when conjugate-matched (for no grating

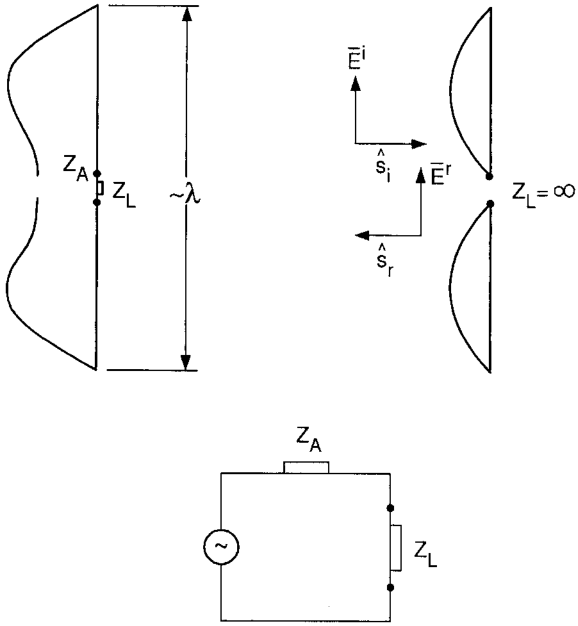


Fig. 2.14 An array of full-wave dipoles without a groundplane will have transmitting and scattering patterns that look alike and scatters the same amount in front and back. However, when $Z_L = \infty$ it will scatter like an array of $\lambda/2$ dipoles; that is, it yields a strong backscatter like a groundplane. Thus, the Thevenin circuit will not predict the correct scattering since no current flows through it. It is therefore not an MSA in the classical sense but it is in the relaxed sense.

lobes). However, for $Z_L = \infty$ it is merely an array of $\lambda/2$ dipoles (see Fig. 2.14), and will consequently exhibit strong scattering in the forward as well as the backward directions. Thus, the Thevenin equivalent circuit does not predict the correct scattering of an array of full-wave dipoles and is thus not an MSA in the classical sense, but it is in the relaxed sense. Furthermore, if backed by a groundplane, we can, as it was the case for the array of $\lambda/2$ dipoles, observe no backscatter for conjugate match (no grating lobes); that is, it is an MSA in the relaxed sense. See also Problem 2.3.

2.11.2 Effect of a Tapered Aperture

A uniform aperture with groundplane can basically absorb all the incident energy and will consequently have no backscatter (see Section 2.6.2). In contrast, a tapered aperture is capable of absorbing only part of the incident energy. Thus, some of the power not being absorbed by Z_L will be either reradiated, most likely in the backscatter direction but not necessarily so, or absorbed by some other absorbing mechanism (see Section 2.13).

This situation is summarized for planar apertures in Fig. 2.15, where we show the uniform aperture at the top and the tapered aperture at the bottom.

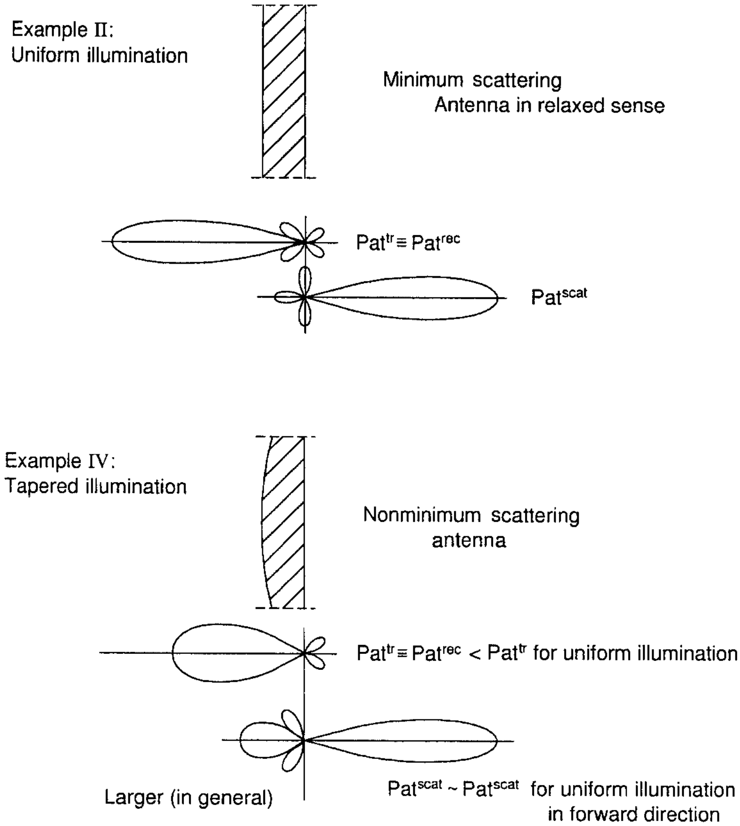


Fig. 2.15 Top: A uniform aperture with a groundplane similar to Example II in Fig. 2.12. Bottom: A tapered aperture will absorb less power than the uniform case, above; however, the scattering in the forward direction will basically be the same. Thus, it scatters more than it absorbs and is thus not an MSA.

We assume that the physical size of the two apertures is the same; that is, the beamwidth of the tapered aperture is somewhat larger and the gain somewhat smaller than for the uniform distribution. Furthermore, the backscatter is larger for the tapered aperture. However, the scattering pattern in the forward direction is essentially the same in the two cases. Thus, while the uniform aperture scatters as much energy as it absorbs, the tapered will scatter more and is consequently not an MSA, not even in the relaxed sense. The discussion above has tacitly assumed that all elements have the same scan impedance. This is not quite the case. A more rigorous and detailed investigation is given in Chapter 5.

The power scattered in the backward sector as a result of the tapered distribution can be absorbed if we work with arrays (see Section 2.13). However, most antennas with tapered aperture distribution are bad candidates for low RCS antennas. Horns, for example, always have a tapered aperture distribution because of the boundary conditions at the walls, and not much can be done about it (see also Chapters 7 and 8). However, an array of horns still has no backscatter for conjugate match, provided that we have no grating lobes (an individual dipole element does not have a "uniform taper" either but is perfect when placed in an array with a groundplane).

2.11.3 The Parabolic Antenna

Typically, a parabolic dish is fed from a horn as shown in Fig. 2.16, top, resulting in a tapered aperture distribution with an aperture efficiency $\eta < 1$. This will produce a transmitting pattern with lower sidelobes, which might be highly desirable for many applications. However, as discussed in the previous section, it will also

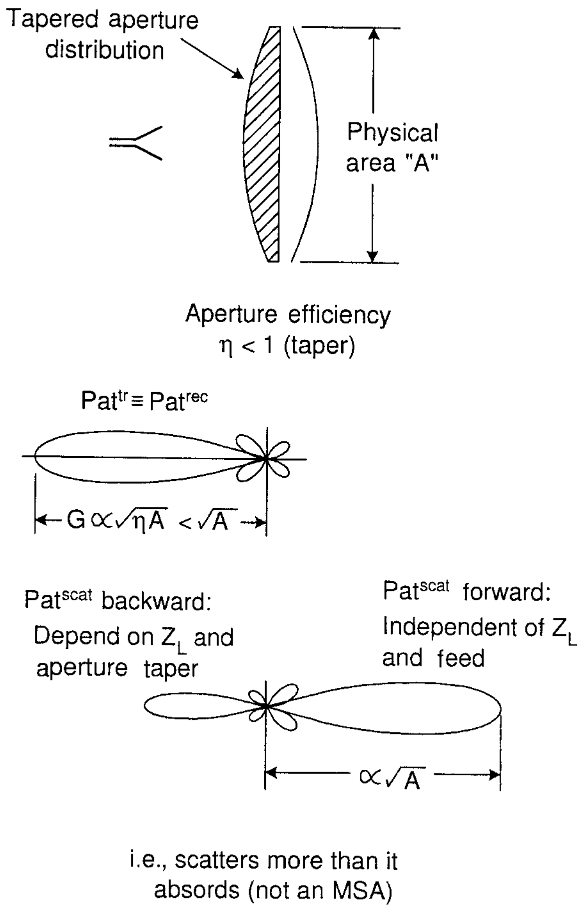


Fig. 2.16 A parabolic antenna has a tapered aperture distribution and will therefore scatter more in the forward scattering direction than it absorbs. Thus, it is not an MSA even in the relaxed sense.

for conjugate match produce an undesirable high level of backscatter; that is, it is not an MSA. However, the highest contribution to the backscatter may actually occur when the incident field is focused on one of the edges of the horn when slightly off boresight as discussed in detail in Chapter 8. Other related subjects will be discussed in Section 2.14.

2.12 HOW TO PREVENT COUPLING BETWEEN THE ELEMENTS THROUGH THE FEED NETWORK

2.12.1 Using Hybrids

The elements in an array will usually be fed via a harness of one sort or another. That opens up the possibility of reradiation via the feed cables unless special precautions are taken. As a simple but very fundamental example, consider Fig. 2.17. Here two subarrays are fed via a simple parallel T connector. Note that the characteristic impedance of the main feed line is Z_0 and how it splits up into two branchlines each with characteristic impedance $2Z_0$ and terminated with the subarrays each with input impedance $2Z_0$. Thus, we will have a perfect match at all frequencies whether we transmit or receive.

In particular, there will be no reradiation from the two subarrays when a signal is incident from broadside (top). However, if a signal is incident at an oblique angle such that the spatial delay between the two subarrays is $(2n + 1)\lambda/2$, the signals from the two subarrays will arrive at the T connector out of phase; that is, they will not continue into the mainline but instead be reflected back to the subarrays where reradiation will take place (bottom). The result is that the backscattering pattern will consist of several very strong lobes (actually there may be other scattering patterns from the subarrays themselves superimposed on these lobes).

We may interpret the backscattering above as caused by a mismatch at the T connector for certain angles of incidence. A very effective way to remedy this dilemma is to use one or more hybrids as shown for example in Fig. 2.18.

Hybrids can be considered as the "fulfillment of the antenna engineer's dream." They come in various types. The one considered here consists of a box with four input ports. When a signal is impressed at port 1, it will emerge at ports 2 and 3 with equal amplitude and 180° out of phase but not at port 4. Similarly, if a signal is impressed at port 2, it will emerge at ports 1 and 4 only, not at port 3. However, if signals of equal amplitude and phase are impressed at ports 2 and 3 simultaneously, a signal will emerge only at port 1, nothing at port 4. Furthermore, if the impressed signals at ports 2 and 3 are equal in amplitude but 180° out of phase, a signal will emerge at port 4 only, nothing at port 1.

From a backscattering point of view, the important feature is that the input impedances at ports 2 and 3 always will be perfectly matched to the subarrays. If the signals from the two subarrays are in phase, a signal will go to port 1—and if out of phase, to port 4—but the input impedances at ports 2 and 3 will always

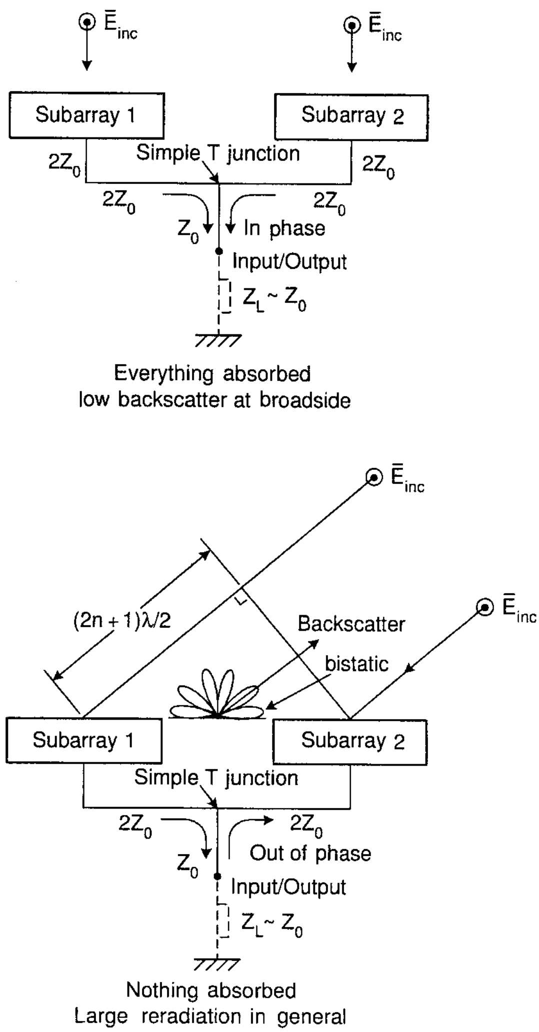


Fig. 2.17 Top: The power to the two subarrays is divided by a simple but perfectly matched T connector. A signal incident at broadside will be perfectly absorbed with basically no backscatter. Bottom: At oblique incidence the two subarrays can be excited out of phase as shown. Thus, the signals from the two subarrays will in that case be reflected at the T connector, resulting in a large backscatter.

be Z_0 regardless of the phase difference between the signals from the subarrays. This is one of the unique features of hybrids.

Thus, all signals incident from any direction will be absorbed by using hybrids resulting in a low backscattering.

The phases of the two subarrays may be controlled by individual phase shifters ϕ_1 and ϕ_2. Still, it has no bearing on the backscatter.

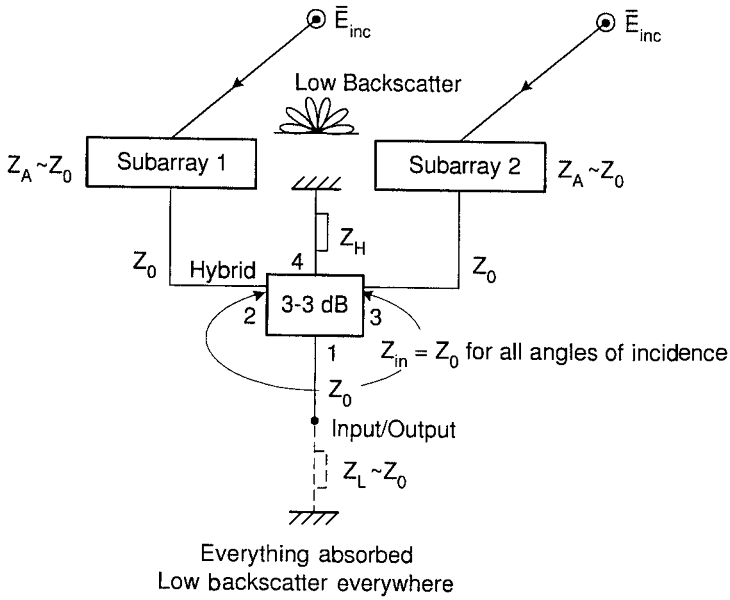


Fig. 2.18 By using a hybrid instead of a T connector as shown above, we can direct the signals from two subarrays into the input impedance when they are in phase and into the hybrid load Z_L when out of phase. In either case (and everything in between) the hybrid input impedance at ports 2 and 3 will be perfectly matched and no significant backscatter will take place.

2.12.2 Using Circulators

Another approach is to use a circulator as shown in Fig. 2.19. Here the transmitter with internal impedance Z_G is connected to port 1, the antenna with antenna impedance Z_A to port 2, and finally the receiver with input impedance Z_R to port 3. The workings of a circulator is now such that a signal applied to port 1 will occur only at port 2, while a signal applied to ports 2 and 3 will occur only on ports 3 and 1, respectively. Thus a signal incident upon the antenna will see only the receiver input impedance Z_R rather than the transmitter impedance Z_G. Since the former in general is matched well to the antenna impedance Z_A, a low reradiation will occur in contrast to seeing Z_G that may be quite different from Z_A (see Section B.9.1).

2.12.3 Using Amplifiers

We saw in the previous section how hybrids and circulators could be applied to prevent coupling between the elements via the feed cables. Another alternative is simply to connect each element to its own individual amplifier. The load impedances then simply become the input impedances of the amplifiers and the isolation from element to element will be high indeed. Furthermore, this approach has the added advantage that the loss in the feed lines can be alleviated by the

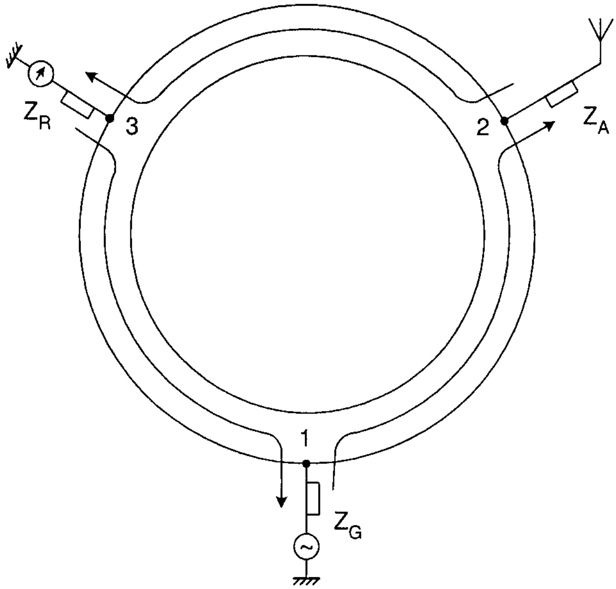


Fig. 2.19 Alternatively we may use circulators as shown above. A generator with impedance Z_G at port 1 delivers a signal only to the antenna Z_A at port 2. However, an incident signal upon the antenna will be directed only to the receiver at port 3. Since in general the receiver impedance Z_R is $\sim Z_A$ while $Z_G \neq Z_A$, we obtain a low backscatter by using a circulator.

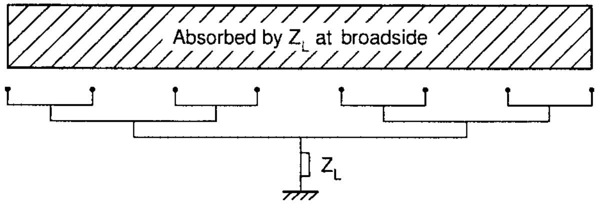
amplifiers if their noise figure is low. Also, we can perform any processing after the receivers or amplifiers without worrying about the backscatter.

2.13 HOW TO ELIMINATE BACKSCATTER DUE TO TAPERED APERTURE ILLUMINATION

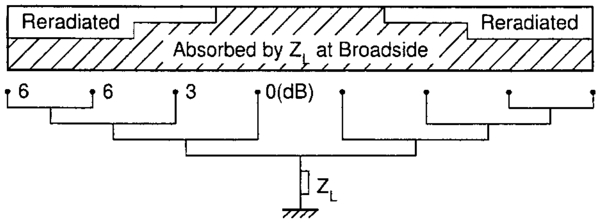
We saw earlier in Section 2.9.2 that a large uniform aperture backed by a groundplane was capable of absorbing an incident plane wave entirely, resulting in very low backscattering. Furthermore, we saw in Section 2.11.2 how a tapered aperture always would backscatter unless special precautions are taken. We will discuss these in this section.

Consider Fig. 2.20, top, where we depict an array where the elements are fed via a simple harness with 3-dB T connectors. Clearly, this will lead to a uniform aperture illumination and consequently to a low backscattering, but only at broadside and certain other angles of incidence as discussed in Section 2.12.1.

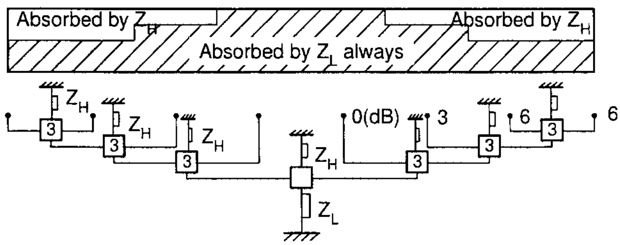
Let us next consider the same array; but as shown in Fig. 2.20, middle, this time we feed the elements via a harness with T connectors arranged to give a tapered aperture illumination as indicated by the cross-hatched area. We will now receive less energy in Z_L while the excess energy between uniform and tapered illumination will be reradiated, resulting in a high backscatter level.



Uniform aperture illumination using simple harness with T connectors. Absorbs everything at broadside and for some oblique angles of incidence



Tapered illumination by using T connectors. Absorbs some at broadside and reradiates some.



Tapered illumination using hybrids. Absorbs everything. No reradiation.

Fig. 2.20 Top: A uniform aperture illumination backed by a groundplane can absorb all the incident energy. Middle: A tapered illuminated aperture can absorb only a part of the incident energy. The other part will be reradiated if we use simple T connectors. Bottom: By substituting the T connectors with hybrids, the excess power can be absorbed in the hybrid loads Z_H rather than reradiated.

We finally in Fig. 2.20, bottom, show the same array as in the middle, but this time we have replaced all the T connectors with the same type of hybrids as used earlier in Fig. 2.18. We will receive the same amount of energy in the load impedance Z_L as in the middle case above; however, the excess energy will be absorbed by the hybrid loads Z_H rather than being reradiated (*Recall*: The antenna sections will always be matched so nothing can be reradiated.).

This conclusion is perhaps best understood if we at first transmit from the main terminal. That will readily produce the tapered aperture distribution as shown. Let us next receive an incident plane wave resulting in equal power delivered

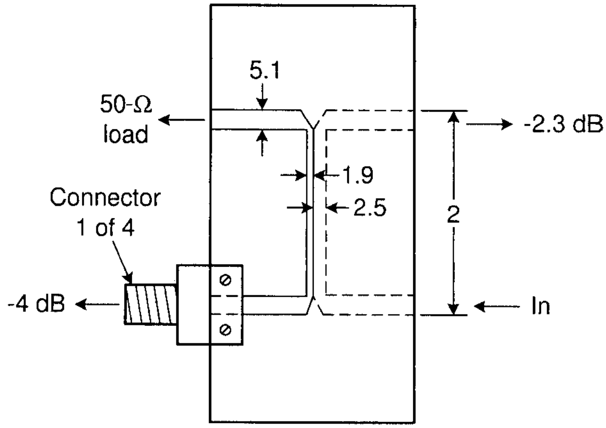


Fig. 2.21 A specially designed hybrid with a power split 4.0/2.3 dB rather than 3/3 dB. All dimensions are in millimeters. (After Larson, private communication).

from each element to the matched input terminals of the hybrids. Thus, no energy can be reflected and reradiated. However, the aperture taper must be the same whether we receive or transmit (reciprocity). This proves our case.

Note: The discussion above assumes that the hybrids are ideal—that is, nothing is being reflected. For a discussion where the hybrid is reflecting a small signal, see reference 59.

The examples above are only typical. Endless variations are possible. For example, the hybrids are not restricted to be merely of the 3-dB type. An example is shown in Fig. 2.21 with a power split of 2.3/4.0 dB. A former Ph.D. student Clayton Larson (now a successful "Big Shot") and the author needed such a hybrid for a very special antenna. Not being an off-the-shelf item, we had to either order one custom-made or make one ourselves. Clay those the latter approach (he was never too keen on telephone negotiations). Overnight he designed, cut the design with an exacto knife and assembled such a hybrid. The next morning we checked it out on the special antenna. It worked flawlessly. Thank you Clay.

2.14 COMMON MISCONCEPTIONS

2.14.1 On Structural Scattering

It has been said several times earlier that what we today prefer to call residual scattering used to be called structural scattering. As one might suspect, this resulted in numerous erroneous ideas. The author remembers in particular one time when he gave a lecture to a group of engineers in industry about RCS of antennas. After the talk one comment was, "We liked your lecture very much and it seems to make a lot of sense. However, we are confused because last week we had a talk from another professor and he told us that the first thing to do to lower the RCS of an antenna is to get rid of the groundplane. You seem to say that the groundplane is our friend. Who is right?"

Well, based on the discussion earlier in Section 2.9, we do of course now know that removing the groundplane simply means that the RCS can get no more than 6 dB below maximum when conjugate-matched. In contrast, when leaving the groundplane in place, the RCS reduction can in principle be infinite! The misconception is of course rooted in the fact that the groundplane constitutes a significant "structure" and the more structure we can get rid of the better! Obviously this is not true.

Another variation on the same theme consists in substituting the groundplane with an absorber (i.e., essentially free space). Well, from an RCS point of view we are of course right back into the 6-dB case (e.g., see Fig. 2.10, left). Furthermore, if you have a transmitting power of 1 kW, you will have about 500 W transformed into heat. That should be enough to fry an omelet.

Then there is of course the case where the groundplane "moves" with frequency. That scam is discussed in Section 6.12.1.3.

2.14.2 On RCS of Horn Antennas

The classical horn antenna has been popular in the microwave community for years. It is therefore only natural that its RCS has been the subject of many investigations. Sadly, it has never emerged as a prime candidate for stealth applications.

This lack of performance is quite often attributed to "structural" scattering. And many designers associate this with scattering from the edges of the horn. Consequently these are "attacked" with all kinds of imaginative treatments ranging from edge cards, serrations, and other loss mechanisms of one kind or another. Although some improvement might be observed in some cases, they must usually be classified as insufficient and often result in some loss of antenna efficiency. The real problem is of course that all horn antennas inherently have a tapered aperture illumination. And as shown in Section 2.11.2, this simply means that a super-low RCS will never emerge for the conjugate-matched classical horn antenna.

However, as also mentioned in Section 2.11.2, this lack of performance of a horn as "soloist" does not prevent it from sounding pretty good in an "orchestra," namely, in an array of horns, as long as no grating lobes are encountered.

For an in-depth discussion of the scattering from horns when used as a feed antenna in a parabolic cylinder, see Chapter 8. There you will also find ways demonstrating how to alleviate some of the problems.

2.14.3 On the Element Pattern: Is It Important?

Based on our discussion in Section 2.9, it should be obvious that the radiation pattern of the individual elements plays almost no role in the array RCS. In fact, as illustrated in Fig. 2.22, top, the total antenna pattern is for no grating lobes so completely dominated by the array factor that any change in the element pattern as indicated by Cases I and II is almost completely drowned out in the total pattern (no grating lobes please—not even close). Thus, the total pattern and consequently the directivity will remain essentially unchanged; that is, we will receive all the incident energy and have no backscatter for uniform aperture illumination (for tapered illumination see Section 2.13). Nevertheless, not everyone is familiar with the content of Section 2.6. Sometimes modifications of the elements are suggested as shown, for example, in Fig. 2.22, bottom. Here to the right we have used slotted waveguides with triangular cross sections rather than the usual square ones shown to the left. Although this idea might look appealing at first glance, the fallacy is of course that the difference in element pattern for the triangular and square cross section is merely of academic interest in this setting. Furthermore, for typical waveguide dimensions required here, the backscatter is not at all as expected from physical optics (a common "sin" committed quite often). At much higher frequencies, this approach makes some sense; however, using a hybrid radome instead will in general be more effective.

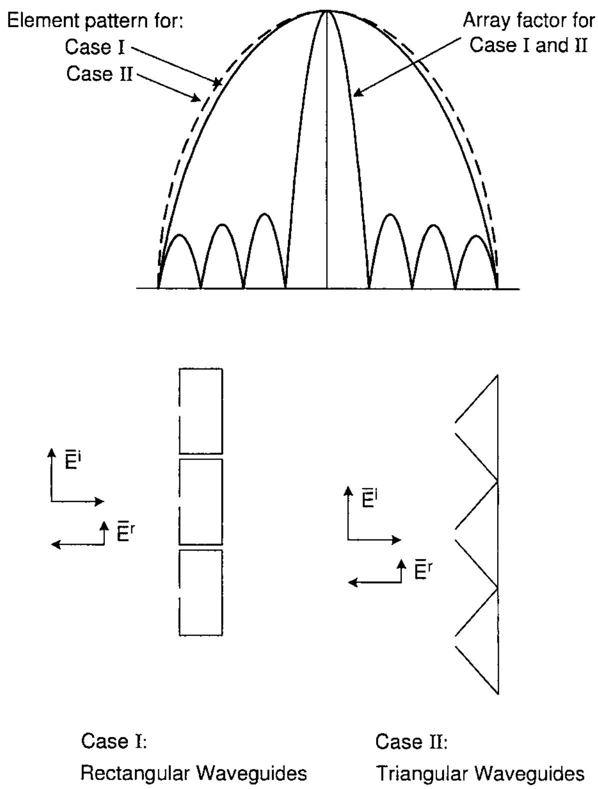


Fig. 2.22 Top: The effect of the element pattern will usually "drown" in the array factor as long as we stay far away from onset of grating lobes. Bottom: Whether you make the array of slotted square waveguides (Case I) or triangular waveguides (Case II), it has very little effect on the scattering in-band. At much higher frequencies, Case II may be superior.

2.14.4 Are Low RCS Antennas Obtained by Fooling Around on the Computer?

It is not unusual to hear the notion that low RCS antennas are obtained by a more or less systematic approach based entirely on computation. This is of course possible. In fact, we addressed this problem in Section 2.3 where we pointed out that any antenna with a residual component $|C| < 1$ could indeed produce a low RCS by simply adjusting the load impedance Z_L in such a way that cancellation between the antenna mode and the residual mode would occur. As also pointed out, this condition is not compatible with maximum power transfer, and perhaps worst of all it is usually narrowbanded, sensitive to the angle of incidence and polarization. Thus, these types of solutions are primarily of academic interest and should not be propagated as anything else when teaching antenna scattering theory on a practical level.

2.14.5 How Much Can We Conclude from the Half-Wave Dipole Array?

Short answer: Not much!

Although it has been stated several times earlier as well as in the summary, we strongly emphasize that the results observed to be true for an array of half-wave dipoles without a groundplane are not necessarily valid for antennas in general. Most notorious is probably the fact that the backscatter for such an antenna when conjugate-matched is reduced 6 dB compared to the maximum return (when short-circuited). This is only true for minimum scattering antennas in the classical sense. If we restrict ourselves merely to consider antennas that do not scatter more than they absorb, we are led for example to arrays of dipoles with a groundplane where for conjugate match we can observe a reduction of ∞ dB. This type is of the MSA type with relaxed conditions. They have certain "flaws;" for example, their Thevenin equivalent circuit does not predict the scattered power for all load conditions. However, they are infinitely more interesting from a practical point of view.

Also, while a dipole array without a groundplane has the same scattering as transmitting pattern, this is far from the case in most of the more interesting cases. See also Chapter 7, Section 7.6.

2.14.6 Do "Small" Antennas Have Lower RCS Than Bigger Ones?

It is well known that the RCS of a large flat plate with physical area A is at broadside proportional to A^2. What is perhaps less well known is the fact that the RCS of the same plate at oblique incidence is basically independent of A (at least in the principal planes). Thus, when it comes to RCS of an object in general, its size may or may not be an issue.

Whatever thought process goes through some people's minds (or lack thereof) it is sometimes suggested that a small antenna in terms of wavelength has a smaller RCS than a larger one. As an example, consider a dipole of total length

$\lambda/2$ and one of length 0.1λ. Now, let there be no doubt that the RCS of the $\lambda/2$-long rod is $\sim 0.8\lambda^2$ whereas it is 5–15 dB lower for the short rod, depending on its wire radius.[1] However, we are of course more interested in the RCS of the two "rods" when they act like dipoles—that is, when they are provided with suitable terminals where we insert a short-circuit for the $\lambda/2$ dipole and a suitable inductance for the 0.1λ dipole that tunes it to resonance. We then merely note that the gains in the two cases are practically the same and consequently according to Section 2.2 will have about the same RCS.

But is there no difference between the two cases at all? Certainly: Namely their bandwidth. The RCS of the short dipole will simply deteriorate faster from its maximum value than will the longer one as a function of frequency. See also Chapter 7.

2.14.7 And the Worst Misconception of All: Omitting the Loads!

We devoted considerable time developing the general scattering theory for antennas, see Section 2.2. We showed that the total RCS of any antenna depends in general strongly on the load impedance Z_L. We further demonstrated these concepts by numerous cases; see, for example, Section 2.9.

Nevertheless, it is not uncommon to encounter presentations that completely ignore the effect of the antenna loads. One paper in particular stands out in my recollection. There the author claimed to have calculated the RCS of a patch antenna. His model was comprised of merely a single patch suspended somewhere in a dielectric slab over an infinite groundplane. No lead-in wire and obviously no loads. When I pointed the deficiency out to him he merely shrugged his shoulders and said, "OK, but this is so hard to do (!)." (Concerning "hard to do," see Appendix D.)

Many other types of antennas have been evaluated typically when their terminals were, for example, short-circuited. This is merely a calculation of an "object," not an antenna (it could of course be part of an important introduction to the antenna problem).

In other words, antennas should not be treated merely as scattering objects with undetermined loads located, for example, in a cavity or elsewhere. The load is extremely important and must always be incorporated to yield meaningful results. Omission of this fact represents in the author's opinion the worst misconception in the entire theory of antenna scattering.

2.15 SUMMARY

We opened this chapter with a review of the classical theory of antenna scattering. The total RCS of any antenna could be written as the phasor addition of two components, namely the *antenna* and the *residual* mode components. The first of these was clearly and precisely defined by the antenna gain G, the reflection

[1] Note that the wire radius affects the bandwidth.

coefficient Γ, and the polarization. The second, however, was more vaguely defined as whatever should be added to the antenna component to obtain the total antenna RCS.

Although this definition sounded somewhat ambiguous, it was nevertheless extremely useful in understanding the complexities of antenna scattering and we were in several important cases able to determine the residual as well as the antenna scattering.

The term residual scattering has also been denoted structural scattering. However, we demonstrated by several examples that this nomenclature was somewhat misleading. We found, for example, that adding a groundplane (i.e., adding structure) to an array of dipoles could reduce the residual scattering to practically nothing rather than increase it. Similarly, a flush-mounted antenna of slots may or may not have any structural scattering even if no structure is visible above the groundplane.

We further considered a very "classical" concept, namely the minimum scattering antenna. Originally it was defined as an antenna that did not scatter more (in total) than it absorbed when conjugate-matched. Furthermore, the scattering pattern should basically be identical to the transmitting pattern with the same amount being scattered in the forward and backward directions. For this class of antennas a Thevenin equivalent circuit would not only predict the correct received power in the load impedance Z_L (always valid) but would also predict the total scattered power as being associated only with the antenna impedance Z_A for all load conditions. Unfortunately, these constraints led to a class of antennas where the RCS for conjugate match is only 6 dB below the maximum RCS (not very impressive indeed).

It was further demonstrated that if we relaxed the conditions above to merely the first, namely that the antenna should scatter no more total energy than it absorbs, it could lead to antennas with virtually no residual scattering (but not necessarily so). An interesting feature of this class of antenna is that the Thevenin equivalent circuit may no longer correctly predict that the total scattered power as merely associated with Z_A under all load conditions. This should be accepted as a fact of life, and failing to realize this can lead to fatal mistakes.

To summarize: By requiring only that the total scattered power does not exceed the absorbed power, we expanded the original minimum scattering antennas with only 6-dB reduction from maximum into a new class of antennas where the reduction could be ∞ dB. Much "respect" has been bestowed upon the classical MSA over the years. However, in this writer's opinion, there is no question what antenna has the real "pedigree."

We also investigated in great detail the effect of aperture illumination upon the RCS of antennas. It was determined that only a uniform aperture illumination was capable of absorbing all the energy incident upon it, resulting in practically no backscatter. If the aperture illumination was tapered, only part of the incident energy could be absorbed. The surplus could be disposed of in basically two different ways. It could be reradiated, resulting in a significant backscatter. Or in case of arrays we could feed each element via appropriate hybrids and lose the excess energy in the hybrid loads, resulting in practically no backscatter. Circulators also serve this purpose.

Alternatively, we could connect each element to its individual amplifier. This would be equivalent to all elements being connected to the same matched load (the input impedances of the amplifiers), resulting in practically no backscatter similar to uniform illumination. The actual aperture tapering or any processing in general could then conveniently be done at the outputs of the amplifiers.

Horn antennas were considered briefly, not so much for their merit as for their imaginative treatment of their edges that seems so fascinating to some. However, as pointed out, the real problem is that horns have an inherent tapered aperture illumination leading to substantial backscatter.

We shall return in much greater detail to feeding parabolic antennas in Chapter 8 and advise alternate ways to alleviate scattering problems. In Chapter 7 we present invisible antennas with omnidirectional radiation patterns.

PROBLEMS

2.1 Consider an array with uniform aperture distribution and assume that the residual scattering as well as edge effects are negligible.

A plane wave is incident upon this array at broadside where it is reflected in the backscattering direction with a reflection coefficient Γ.

Find the upper limits of the VSWR as seen at the array terminals if we require the reflection coefficient magnitude $|\Gamma|$ to be less than

1. 10 dB
2. 26 dB
3. 32 dB
4. 40 dB

This is a very easy, but important, problem.

See also Sections 5.8 and 5.9.

2.2 Show that

$$\sqrt{\sigma_{ant}} = \tfrac{1}{2}(\sqrt{\sigma_{tot\ max}} - \sqrt{\sigma_{tot\ min}})$$

$$\sqrt{\sigma_{res}} = \tfrac{1}{2}(\sqrt{\sigma_{tot\ max}} + \sqrt{\sigma_{tot\ min}})$$

These equations can be very helpful in determining σ_{ant} and σ_{res} experimentally.

2.3 Consider an array of full-wave dipoles without a groundplane rather than half-wave dipoles as shown in Fig. 2.8.

Find the component C associated with the residual scattering and show that this array obeys the 6-dB Rule (approximately).

You may assume that $\Gamma_{FSS} = -1$ when open-circuited and $\Gamma_{FSS} \sim 0$ when short-circuited, The last approximation actually becomes basically exact at a frequency somewhat higher than the full-wave frequency (normal angle of incidence). Does the Thevenin equivalent circuit hold for all load conditions?

2.4 Consider an infinite array of half-wave dipoles with a groundplane at arbitrary distance as shown in Fig. 2.9.

Show that the sum of the three plane waves marked "1," "2," and "3" does indeed add up to zero for conjugate match.

To the instructor: This is a somewhat lengthy problem that should be assigned only as punishment! In fact, it is much simpler to use the equivalent circuit shown in Fig. 2.10, right.

2.5 Consider the equivalent circuit for an array of half-wave dipoles without a groundplane as shown in Fig. 2.10, left.

Show that the signal transmitted in the forward direction as well as the reflected signal are both 6 dB below the incident signal when conjugate-matched.

Also verify the conservation of power.

2.6 Single Dipole with Parasitic Dipole Reflector.

In Fig. P2.6 we show a single dipole with self-impedance $Z_{1,1}$ backed by a single parasitic element with self-impedance $Z_{2,2}$ and load impedance $Z_L^{(2)}$. Their mutual impedances are denoted $Z_{1,2} = Z_{2,1}$.

You are asked to find the transmitting, receiving, and scattering patterns for this antenna configuration.

Transmitting Case: The active element is fed from a voltage generator V_g as shown in Fig. P2.6, top. The element currents under transmitting condition are denoted $I_t^{(1)}$ and $I_t^{(2)}$ for the active and parasitic element, respectively.

The generalized Ohm's Law for the system then becomes

$$V_g = Z_{1,1} I_t^{(1)} + Z_{1,2} I_t^{(2)}$$
$$0 = Z_{2,1} I_t^{(1)} + (Z_{2,2} + Z_L^{(2)}) I_t^{(2)}$$

a. Find the input impedance $Z_{in} = \dfrac{V_g}{I_t^{(1)}}$ for the antenna system.

b. Find the relative transmit pattern Pat^{tr} as indicated in Fig. P2.6, top, expressed by $I_t^{(1)}$, $Z_{2,1}$, $Z_{2,2} + Z_L^{(2)}$ and the distance d between the two elements. Sketch the transmit pattern Pat^{tr} based on your expression.

Receiving Case: We now remove the generator V_g from the antenna terminals and expose it instead to an incident plane wave $\overline{E}^i$ with direction of propagation $\hat{s}$ as shown in Fig. P2.6, bottom. The terminals are loaded with the load impedance $Z_L^{(1)}$ and the induced voltages in the active and passive element are called $V^{(1)}$ and $V^{(2)}$, respectively (we need not actually

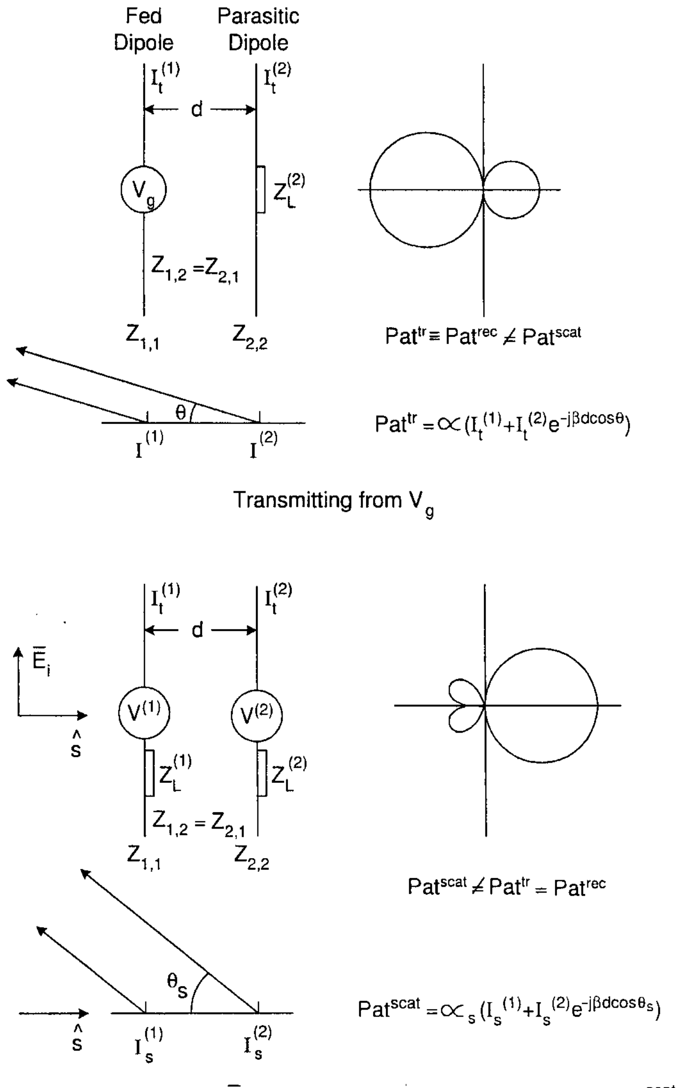


Fig. P2.6 Determination of the transmitting, receiving, and scattering patterns for an antenna comprised of one driven element (or column) and one parasitic element (or column).

calculate these voltages but in case you are curious; see reference 60). Thus, the generalized Ohm's Law for the receiving case becomes

$$V^{(1)} = (Z_{1,1} + Z_L^{(1)})I_s^{(1)} + Z_{1,2}I_s^{(2)}$$
$$V^{(2)} = Z_{2,1}I_s^{(1)} + (Z_{2,2} + Z_L^{(2)})I_s^{(2)}$$

where $I_s^{(1)}$ and $I_s^{(2)}$ denote the element currents under scattering (receiving) conditions.

c. You may now assume that the length of the two elements are the same (we can always tune the parasitic element by adjusting $Z_L^{(2)}$). Thus,

$$V^{(2)} = V^{(1)} e^{-j\beta d}$$

Find the relative scattering pattern Pat^{scat} as expressed by $V^{(1)}$ and all the appropriate impedance terms as well as the bistatic angle θ_s and the element spacing d. Sketch the pattern Pat^{scat} based on your expression.

d. Find the load impedance $Z_{L\ BS}^{(1)}$ that produces zero backscattering.

e. Find the load impedance $Z_{L\ max}^{(1)}$ that will absorb most of the incident power. Compare $Z_{L\ BS}^{(1)}$ to $Z_{L\ max}^{(1)}$ and estimate the power loss by using $Z_{L\ BS}^{(1)}$ instead of $Z_{L\ max}^{(1)}$.

f. Which one of the two patterns above (if any!) represents the receiving pattern Pat^{rec}?

2.7 Consider an array of four elements with individual reflectors fed from a corporate feed network as shown in Fig. P2.7. The two subarrays are fed via two simple T connectors with characteristic impedances as shown. However, the two T connectors are fed from an ideal hybrid as also shown. An incident plane wave $\overline{E}^i$ is incident upon this array at an angle η. Due to mismatch between the antenna impedance Z_A and the feed cables, we will

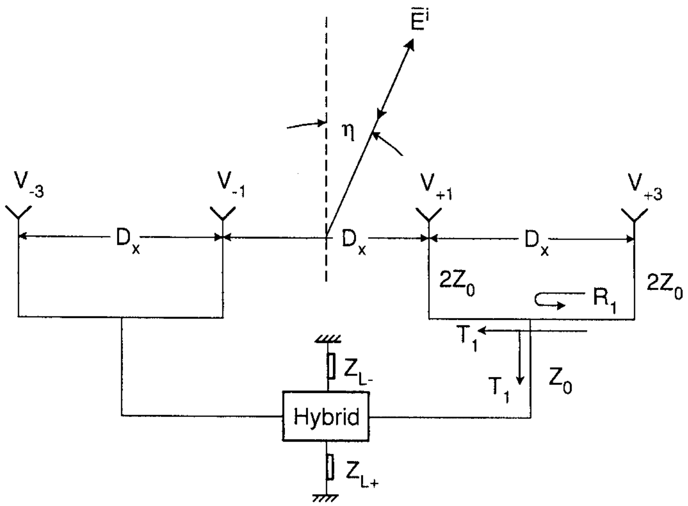


Fig. P2.7 A simple array of four elements fed via a harness comprised of two simple T connectors fed via a simple hybrid.

experience a mainbeam in the bistatic direction and some smaller sidelobes in the backscatter directions. However, you are only required to find the scattering pattern due to reflections from the two T connectors at oblique angle of incidence η. Estimate the magnitude of the backscattered field.

Hint: In general,

$$G = \frac{4\pi}{\lambda^2} A,$$

that is, (2.8) can for $G_i = G_r$ and $p_i = p_r = 1$ be written as

$$\sigma_{ant} = \frac{4\pi}{\lambda^2} A^2 \Gamma^2$$

where $\frac{4\pi}{\lambda^2} A^2$ is the RCS of a flat plate with physical area A.